Doppler Compensation for Underwater Acoustic Links

Who removes which Doppler, where and when — for group discussion

9 October 2026

Target: IEEE JSAC special issue on Next-Generation Underwater Acoustic Communication Systems (deadline 31 March 2027)

CLAIM to be shown SIMULATED simulation result SUGGESTED design proposal OPEN needs material, a proof or an experiment SOURCE found by search, not yet checked in full

Main idea

The ground rule: a great coffee machine, given good coffee

Contribution: the compensation — where, when and how a known Doppler is removed, how much of it remains, at what cost.

Taken as given: the estimation. It enters only as an input model:

- own velocity from an IMU: error σ_{imu}, no delay
- slow part (geometry, drift, path direction u) from the link: error σ_{slow}, age Δ
- residual inside the packet, at the receiver: error σ_{rx}

No new estimator is developed.

Doppler of a link

$$\epsilon(t) = \frac{\big(v_A(t) - v_B(t)\big) \cdot u(t)}{c}$$

v_A, v_B: transducer velocities
u: direction of the path

fast part: wave-driven motion of each node
slow part: drift, cruise, geometry

Phase 0: a good estimate goes stale in about a second

Real buoy motion	2.3–2.4 s ahead	share $<$10 s waves
CDIP Waverider (0.9 m), 3 Pacific sites	0.77–1.08	45–95%
OOI surface buoys (3.3 m, 4 t), Oregon, Irminger	0.76–0.93	70–94%

Normalised error of the best linear prediction of velocity (1 = no better than the mean), from noisy per-packet measurements every 2.3–2.4 s; swell, mixed and wind-sea days. Every 0.8 s, 0.8 s ahead: 0.27–0.47.

Consequences

- the far end's fast motion cannot cross a round trip (2–3 s at 2 km)
- fast motion must be compensated where it is measured, when it is measured
- only the slow part can be shared and predicted

Hs from the reconstructed motion matches each buoy's own wave sensor within 1–5%. Details: `code/PLAN.md`.

The proposal: split compensation SUGGESTED

Each Doppler component is removed by the node that knows it:

1. the **transmitter** removes its own motion, measured live, while it emits
2. the **receiver** removes its own motion while it receives
3. the **slow part** is exchanged over the link and predicted
4. the **receiver** removes what remains, inside the packet

Candidate claim SUGGESTED: decide, for each component, *who* removes it and *how* (trajectory resampling, frequency correction, per-user branches), from the accuracy and age of the estimate at that node; with closed-form residuals and real motion data, on single links and multi-user up- and downlinks.

Scope: OFDM; mobile or surface-coupled nodes; one link, or K nodes and a gateway. Bulk Doppler resampled; path-specific Doppler counted as residual.

Gaps with the literature

Related work (1): compensating a time-varying Doppler in the receiver

Work	Compensation	Limit for us
Sharif et al., JOE 2000	Resampling with velocity and acceleration	One trajectory per packet
Li et al., JOE 2008	Resampling + residual frequency offset (nonuniform Doppler)	Constant scale per block
Zakharov et al., JOE 2019	Packet designed for a motion range	Fixed offline
Abdelkareem et al., Ad Hoc Netw. 2016	Time-varying Doppler, previous packet gates the search	Receiver only
Yoshizawa et al., 2018	Parallel resampling for fluctuating Doppler	Receiver only
Cho–Kim, IET Commun. 2026	Constant acceleration from the cyclic prefix	$\approx$1 s frames; jerk ignored

All compensate at the receiver, after reception; compared by BER in chosen scenarios. Analyses exist for a constant-scale error (Mason 2008) and thresholds (rate $< 10^{-5}$, Stojanovic 2006; scale spread $< 10^{-3}$, Yerramalli–Mitra 2011).

Related work (2): several Doppler, transmit side, sensors

Work	Mechanism	What it lacks
Tu, Duman, Stojanovic, Proakis 2011, 2013	OFDMA; one resampling branch per user at the receiver	K estimated branches; interference remains
Daoud et al., IEEE TVT 2015	Multiple resampling, multiple-access channel	Receiver only
Hasan et al., Ad Hoc Netw. 2025	Guard bands adapted to mobility	Not to the residual
Oiwa et al., IEICE Tech. Rep. 2020	Accelerometer-aided compensation	One end; not peer reviewed
Choi et al. 2022; Aubry 2019	Modem Doppler into navigation filters	Not used to compensate
3GPP NTN (TS 38.300)	UE pre-compensates Doppler before sending	From ephemeris
Cuji Dután 2025 (thesis)	Stale feedback at the transmitter (beamforming)	Solved with slow angles

Library, web, Scholar and 159 citing works checked: no UWA transmitter compensates Doppler. Say "transmit-side": UWA papers use "Doppler pre-compensation" for a receiver step.

Candidate gaps

G1 Partly open: no design rule for intra-packet compensation

Constant-scale error (Mason 2008) and ICI of scaling or linear drift (Stojanovic et al. 2012; radio closed forms) are analysed; no rule combines estimate errors, model order, packet length and real motion statistics. Supporting result.

G2 No UWA transmitter compensates Doppler; placement vs estimate age not analysed

Only NTN pre-compensates at the transmitter, from ephemeris. Phase 0 gives the age limits.

G3 Several Doppler handled only at the receiver

One estimated branch per user (Tu et al. 2011, 2013); no split between users and gateway; guard bands follow mobility (Hasan 2025).

G4 Sensors drive navigation, not compensation

Only Oiwa et al. 2020, at one end, in a technical report.

Proposed methodology

C1: compensation inside the packet (receiver) SUGGESTED

Resample along an estimated trajectory $\hat{\epsilon}(t)$, then correct the frequency, in baseband:

$$r(t) \approx s\Big(t + \int_0^t \epsilon\Big) \;\longrightarrow\; s\Big(t + \int_0^t (\epsilon - \hat{\epsilon})\Big)$$

Trajectory models: constant, linear, quadratic, piecewise.

Output: residual interference between subcarriers and SINR loss vs estimate error, model order, packet length; complexity.

Baselines

- one scale per packet (pre/postamble)
- Cho–Kim: constant acceleration
- per-symbol correction

Theory target (R1): closed-form residual; design rule for the model order given packet length and motion.

For an 8 s swell, a constant-acceleration fit leaves 3% of a subcarrier spacing over 0.9 s, 46% over 4 s (`literature/baseline_cho_kim_2026.md`).

C2: where to compensate — transmitter, receiver or split SUGGESTED

Split compensation

- transmitter: own motion projected on u, measured live, applied while emitting (real-time resampling of the outgoing waveform)
- receiver: own motion, live, applied while receiving
- slow part: exchanged and predicted (age Δ tolerable)
- remainder: C1 at the receiver

Compared with receiver-only compensation, vs estimate age, sea state, packet length; measured buoy motion as input.

Theory target (R2): staleness

$$\mathbb{E}\left[(v(t+H) - v(t))^2\right]$$

$$= 2\int S_v(f)\left(1 - \cos 2\pi f H\right) df$$

and its prediction counterpart: when transmit-side, receive-side or split compensation is best.

IMU error, latency, lever arm and accuracy of u: now modelled and swept (C2 results).

C3: several Doppler at once SUGGESTED

Uplink, K nodes to a gateway

- each node removes its own motion before sending (C2)
- users then differ only by the gateway's motion projected on their directions, and by slow terms
- a moving gateway still needs one branch per user for its own motion, but set from its IMU and u_k, not estimated; users in similar directions can share a branch

Downlink: the gateway knows its own motion live, so it pre-compensates each user's subband before sending, with no prediction.

Guard bands sized by the residual after compensation, not by the raw mobility.

Baseline: multiple-resampling receiver (Tu et al. 2011, 2013), contiguous and interleaved allocation.

Theory target (R3): interference between users vs residuals and guard bands; users supported vs number of branches and per-user estimation.

Evaluation

Inputs

- motion: measured buoy records (CDIP, OOI) replayed; AUV and seabed nodes from models
- channel: Bellhop geometry along the trajectories (binary in the DESERT build); Channel Library records if needed
- estimates: true Doppler + error with the stated σ and age

Two levels

- waveform simulation (Python): residual, interference, BER/PER, complexity
- network level (DESERT Doppler extension): many nodes, MAC, traffic, with the residual tables from the waveform level

Code plan: `code/PLAN.md`.

C2 results (preliminary)

C2: split wins on all three links SIMULATED

SIR between subcarriers [dB]	Buoy→ seabed	Buoy– buoy	AUV→ buoy
Receiver, previous packet's estimate	−5 to 6	−5 to 2	−4 to 4
Receiver, quadratic, 1 s packet	≤36	≤34	≤35
Receiver, per symbol	27–36	24–33	26–35
One end's IMU only	30–34	−4 to 7	−2 to 7
Both IMUs, no receiver estimate	—	28–29	28–30
Split: both IMUs + receiver line from 2 estimates	**37–41**	**32–38**	**33–40**

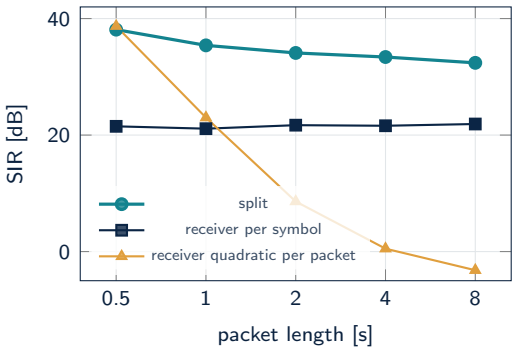
Three days (CDIP swell, CDIP wind sea, OOI Irminger swell), 1 and 4 s packets.

15 sea-state days: split best in all 30 cases, median 9 dB above the per-symbol receiver.

Setting

- OFDM: 16 kHz carrier, 8 kHz band, 1024 subcarriers
- real buoy motion (CDIP, OOI with gyro); AUV from a model
- IMU drift 0.01 m/s, 50 ms latency bridged by the accelerometer, lever arm corrected
- receiver estimate error $3 \cdot 10^{-6}$

Longer packets favour split; stale estimates fail at any range SIMULATED



Buoy–buoy, OOI Irminger swell, 1 km.

- below ≈0.5 s the receiver alone matches split; beyond, one model per packet cannot follow the waves
- split keeps 32–38 dB up to 8 s: the IMUs remove the waves, the receiver line only the slow rest
- **range** 0.5–4 km: split 35–37 dB at every range; the estimate from the previous round trip (1.7–6.3 s old) gives −12 to −5 dB — Phase 0 confirmed on links

What split needs from the sensors SIMULATED

Factor	Split	IMU only
IMU drift 0.005 → 0.1 m/s	38–41 → 28–32	29–39 → 11
Latency 0 / 50 / 100 / 200 ms	42 / 37 / 30 / 20	32 / 33 / 28 / 16
50 ms, not extrapolated	25	17
Lever arm 2 m, corrected / not	38 / 13	31 / 3
Path direction error 0 / 5 / 20°	37 / 30 / 18	30 / 13 / 0

dB; buoy→seabed on the OOI buoy (direction: buoy–buoy gives 37 / 26 / 15 for split).

Requirements

- an IMU about ten times worse is tolerated: the receiver line removes its slow drift
- latency below ≈100 ms, bridged by the accelerometer
- lever-arm correction from the gyro is mandatory on a large buoy
- path direction within ≈5° (IMU alone: ≈2°, the 4 t buoy moves 1 m/s across the path)

A floor no endpoint can remove: surface paths with their own Doppler SIMULATED

Channel	Seabed CDIP	Seabed OOI	Buoy–buoy
single path (split)	34	33	33
5 paths, common Doppler	34	31	30
+ surface-path Doppler (all strategies)	11–12	4	4
closed-form floor	14	4	—

dB, 1 s packets, receiver estimating the channel from pilots (ceiling ≈40 dB).

Each surface bounce adds $2 \sin \theta_k \, w(t)/c$ (w: vertical velocity of the surface). Interference floor:

$$\sum_k p_k \left(\tfrac{2 \sin \theta_k}{c}\right)^2 \mathrm{var}(w)$$

- placement matters only above this floor
- our model is a worst case: mirror-like bounces, the buoy's full heave, 40% of the power
- several Doppler on one link: the motivation for C3

Toward the placement rule SUGGESTED

Evaluator: for every placement (IMU stage at each end on/off $\times$ receiver stage: none, per symbol, constant or line from 2 estimates, quadratic from all), the interference from

- motion spectra through each node's stage (latency, lever arm, direction, with cross terms)
- the fit the receiver really makes (line through 2 estimates: 2.7$\times$ the best line's error)
- the phase within each FFT window (a residual ramp: 5$\times$ less interference than an offset)

SIMULATED 85 points: same winner at 81 (misses $\leq$0.6 dB); all strategies within $\pm$0.7 dB (median).

First reading of the rule

- a node removes its own fast motion if its latency, bridged, is $\lesssim$0.01 wave periods
- a static node runs no IMU stage (it only adds drift)
- the receiver's role depends on its estimate budget: 2 per packet $\rightarrow$ split; one per symbol $\rightarrow$ IMU + per-symbol receiver, predicted higher

OPEN maps for both budgets (packet, latency, IMU quality in wave units); new placements being simulated.

Theoretical objectives and risks

Theory: three results

R1 Inside the packet

Residual interference and SINR loss vs estimate error, trajectory model order, packet length; design rule for the model order

R2 Staleness and placement

Compensation error vs age of the estimate, from the motion spectrum; when transmit, receive or split compensation is best; validated on buoy data

R3 Several Doppler

Interference between users vs residuals and guard bands; users supported vs branches and per-user estimation

Weight: R2 carries the main message and rests on real data; R1 and R3 make it a design. R1 and R2 share one tool SUGGESTED: errors of representing $v(t)$ (old sample; polynomial over the packet) as integrals of the motion spectrum. Closed forms for R1 and R2 now match the simulation (C2 results); OPEN not yet proved or written up.

Risks

Compensation is a classic topic. Novelty must lie in placement and splitting (G2, G3), backed by the staleness analysis (R2), not in a new resampler.

Sensors. IMU error, lever arm and path direction enter the transmit-side correction; their accuracy must be modelled. "Why not an IMU?" becomes part of the design.

Path-specific Doppler cannot be removed by one resampling per link. Simulated: surface paths with their own Doppler set a floor for every strategy (C2 results); placement matters only above it.

Data. Motion is real; channel and multi-node geometry are simulated. AUV and seabed scenarios need models or logs.

OPEN Search for split or sensor-driven compensation beyond the local library; read Mason 2008 and Tanash 2026 before R1.

Questions for the group

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- Decided: scenarios buoy $\rightarrow$ seabed, buoy–buoy, buoy–AUV; a node's own IMU counts as a good estimate; SIR now, coded BER/PER later.
- Can the modem resample the outgoing waveform in real time from the IMU, or only per packet?
- How many Doppler estimates per packet can the receiver afford (preamble/postamble only, or pilots in every symbol)?
- Is the downlink (gateway pre-compensating each user) in scope for the first paper?